

# ML-DL-Ops Assignment 5

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## Objective

This assignment is divided into two parts. Q1 involves fine-tuning a pre-trained ViT-Small model on CIFAR-100 — first without LoRA, then with LoRA using PEFT for various rank and alpha combinations, followed by hyperparameter optimization using Optuna. Q2 focuses on adversarial attacks using IBM Adversarial Robustness Toolbox (ART) on CIFAR-10 with ResNet-34, covering FGSM (from scratch and via ART) and adversarial detection using PGD and BIM attacks. All experiments were conducted inside a Docker container.

## Q1: ViT-S Fine-tuning on CIFAR-100 with LoRA

### Setup & Model

- **Base Model:** ViT-Small (ViT-S/16) pre-trained on ImageNet via `timm`
- **Dataset:** CIFAR-100 (50,000 train / 10,000 test, 100 classes)
- **Classification Head:** Replaced with a linear layer (`hidden_dim`  $\rightarrow$  100)
- **Training:** 10 epochs, AdamW optimizer, CosineAnnealingLR scheduler
- **LoRA via PEFT:** Injected into Q, K, V attention projection weights
- **LoRA Dropout:** Fixed at 0.1 for all experiments

### Experiment 0 (No LoRA): Fine-tune Classification Head Only

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 1.1430     | 0.7910   | 70.61         | 77.08       |
| 2     | 0.6761     | 0.7334   | 80.29         | 78.47       |
| 3     | 0.5899     | 0.7356   | 82.66         | 78.49       |
| 4     | 0.5364     | 0.7144   | 84.13         | 79.15       |
| 5     | 0.4936     | 0.7196   | 85.34         | 78.95       |
| 6     | 0.4604     | 0.7122   | 86.28         | 78.88       |
| 7     | 0.4352     | 0.7108   | 87.07         | 79.45       |
| 8     | 0.4142     | 0.7090   | 87.84         | 79.54       |
| 9     | 0.3987     | 0.7059   | 88.38         | 79.39       |
| 10    | 0.3902     | 0.7045   | 88.75         | 79.48       |

Experiment 0 — No LoRA, classification head only

### LoRA Experiments: Train-Val Tables per Configuration

The following table template is repeated for all 9 rank–alpha combinations ( $\text{Rank} \in \{2, 4, 8\}$ ,  $\text{Alpha} \in \{2, 4, 8\}$ ,  $\text{Dropout} = 0.1$ ):

**Experiment 1 — Rank: 2, Alpha: 2, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6810     | 0.3821   | 82.00         | 87.98       |
| 2     | 0.2907     | 0.3591   | 90.89         | 88.92       |
| 3     | 0.2186     | 0.3606   | 92.97         | 89.18       |
| 4     | 0.1682     | 0.3537   | 94.59         | 89.57       |
| 5     | 0.1315     | 0.3673   | 95.95         | 89.38       |
| 6     | 0.1043     | 0.3647   | 96.90         | 89.69       |
| 7     | 0.0840     | 0.3654   | 97.75         | 89.62       |
| 8     | 0.0704     | 0.3697   | 98.28         | 89.68       |
| 9     | 0.0620     | 0.3710   | 98.65         | 89.76       |
| 10    | 0.0578     | 0.3712   | 98.83         | 89.78       |

**Experiment 2 — Rank: 2, Alpha: 4, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6799     | 0.3716   | 82.05         | 88.67       |
| 2     | 0.2859     | 0.3568   | 91.07         | 88.98       |
| 3     | 0.2147     | 0.3640   | 93.12         | 88.96       |
| 4     | 0.1637     | 0.3540   | 94.79         | 89.34       |
| 5     | 0.1267     | 0.3620   | 95.99         | 89.57       |
| 6     | 0.0968     | 0.3643   | 97.17         | 89.59       |
| 7     | 0.0761     | 0.3641   | 98.04         | 89.76       |
| 8     | 0.0629     | 0.3679   | 98.51         | 89.55       |
| 9     | 0.0539     | 0.3692   | 98.92         | 89.64       |
| 10    | 0.0491     | 0.3692   | 99.08         | 89.67       |

**Experiment 3 — Rank: 2, Alpha: 8, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6529     | 0.3908   | 82.63         | 87.77       |
| 2     | 0.2847     | 0.3645   | 90.93         | 88.81       |
| 3     | 0.2130     | 0.3664   | 93.08         | 89.12       |
| 4     | 0.1607     | 0.3662   | 94.83         | 89.55       |
| 5     | 0.1208     | 0.3723   | 96.22         | 89.38       |
| 6     | 0.0904     | 0.3799   | 97.44         | 89.29       |
| 7     | 0.0691     | 0.3777   | 98.14         | 89.55       |
| 8     | 0.0552     | 0.3802   | 98.78         | 89.70       |
| 9     | 0.0465     | 0.3821   | 99.08         | 89.69       |
| 10    | 0.0416     | 0.3830   | 99.27         | 89.70       |

**Experiment 4 — Rank: 4, Alpha: 2, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6847     | 0.3869   | 82.06         | 88.02       |
| 2     | 0.2871     | 0.3464   | 90.86         | 89.30       |
| 3     | 0.2127     | 0.3569   | 93.10         | 89.21       |
| 4     | 0.1642     | 0.3496   | 94.74         | 90.04       |
| 5     | 0.1235     | 0.3523   | 96.14         | 90.02       |
| 6     | 0.0951     | 0.3588   | 97.20         | 89.93       |
| 7     | 0.0743     | 0.3619   | 98.06         | 90.27       |
| 8     | 0.0623     | 0.3644   | 98.57         | 90.22       |
| 9     | 0.0534     | 0.3665   | 98.89         | 90.17       |
| 10    | 0.0501     | 0.3665   | 99.03         | 90.13       |

**Experiment 5 — Rank: 4, Alpha: 4, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6658     | 0.3742   | 82.48         | 88.56       |
| 2     | 0.2823     | 0.3570   | 90.93         | 89.12       |
| 3     | 0.2040     | 0.3570   | 93.29         | 89.07       |
| 4     | 0.1519     | 0.3541   | 95.09         | 89.86       |
| 5     | 0.1110     | 0.3649   | 96.48         | 89.67       |
| 6     | 0.0816     | 0.3745   | 97.66         | 89.54       |
| 7     | 0.0608     | 0.3716   | 98.49         | 89.70       |
| 8     | 0.0480     | 0.3788   | 99.02         | 89.86       |
| 9     | 0.0407     | 0.3783   | 99.29         | 90.00       |
| 10    | 0.0367     | 0.3793   | 99.40         | 89.90       |

**Experiment 6 — Rank: 4, Alpha: 8, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6552     | 0.3767   | 82.61         | 88.40       |
| 2     | 0.2750     | 0.3480   | 91.19         | 89.32       |
| 3     | 0.1941     | 0.3639   | 93.77         | 89.30       |
| 4     | 0.1385     | 0.3671   | 95.61         | 89.41       |
| 5     | 0.0977     | 0.3759   | 97.04         | 89.70       |
| 6     | 0.0691     | 0.3787   | 98.11         | 89.88       |
| 7     | 0.0494     | 0.3824   | 98.81         | 89.99       |
| 8     | 0.0367     | 0.3867   | 99.33         | 90.00       |
| 9     | 0.0302     | 0.3872   | 99.52         | 90.11       |
| 10    | 0.0269     | 0.3877   | 99.66         | 90.10       |

**Experiment 7 — Rank: 8, Alpha: 2, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6850     | 0.3734   | 81.95         | 88.38       |
| 2     | 0.2844     | 0.3578   | 91.14         | 89.10       |
| 3     | 0.2106     | 0.3400   | 93.21         | 89.84       |
| 4     | 0.1575     | 0.3577   | 94.94         | 89.42       |
| 5     | 0.1189     | 0.3595   | 96.35         | 89.69       |
| 6     | 0.0896     | 0.3665   | 97.43         | 89.56       |
| 7     | 0.0687     | 0.3685   | 98.27         | 89.68       |
| 8     | 0.0560     | 0.3726   | 98.73         | 89.70       |
| 9     | 0.0479     | 0.3724   | 99.04         | 89.82       |
| 10    | 0.0447     | 0.3726   | 99.17         | 89.71       |

**Experiment 8 — Rank: 8, Alpha: 4, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6755     | 0.3708   | 82.12         | 88.41       |
| 2     | 0.2748     | 0.3463   | 91.29         | 89.15       |
| 3     | 0.1977     | 0.3603   | 93.67         | 89.18       |
| 4     | 0.1409     | 0.3537   | 95.54         | 89.73       |
| 5     | 0.0986     | 0.3603   | 97.04         | 89.76       |
| 6     | 0.0707     | 0.3692   | 98.12         | 89.98       |
| 7     | 0.0516     | 0.3761   | 98.81         | 89.79       |
| 8     | 0.0396     | 0.3789   | 99.26         | 89.73       |
| 9     | 0.0331     | 0.3805   | 99.45         | 89.78       |
| 10    | 0.0299     | 0.3807   | 99.57         | 89.96       |

**Experiment 9 — Rank: 8, Alpha: 8, Dropout: 0.1**

| Epoch | Train Loss | Val Loss | Train Acc (%) | Val Acc (%) |
|-------|------------|----------|---------------|-------------|
| 1     | 0.6556     | 0.3721   | 82.49         | 88.79       |
| 2     | 0.2693     | 0.3479   | 91.44         | 89.52       |
| 3     | 0.1844     | 0.3574   | 94.13         | 89.79       |
| 4     | 0.1267     | 0.3657   | 96.00         | 89.89       |
| 5     | 0.0846     | 0.3725   | 97.48         | 90.06       |
| 6     | 0.0553     | 0.3883   | 98.56         | 90.01       |
| 7     | 0.0370     | 0.3898   | 99.20         | 90.04       |
| 8     | 0.0365     | 0.3893   | 99.55         | 90.25       |
| 9     | 0.0210     | 0.3926   | 99.71         | 90.20       |
| 10    | 0.0182     | 0.3993   | 99.78         | 90.24       |

**Test Accuracy Summary Table**

| LoRA Layers              | Rank     | Alpha    | Dropout    | Overall Test Acc (%) | Trainable Params |
|--------------------------|----------|----------|------------|----------------------|------------------|
| Without LoRA (head only) | –        | –        | –          | 79.54                | 38500            |
| Q, K, V                  | 2        | 2        | 0.1        | 89.78                | 75364            |
| Q, K, V                  | 2        | 4        | 0.1        | 89.76                | 75364            |
| Q, K, V                  | 2        | 8        | 0.1        | 89.70                | 75364            |
| Q, K, V                  | <b>4</b> | <b>2</b> | <b>0.1</b> | <b>90.27</b>         | <b>112228</b>    |
| Q, K, V                  | 4        | 4        | 0.1        | 90.00                | 112228           |
| Q, K, V                  | 4        | 8        | 0.1        | 90.11                | 112228           |
| Q, K, V                  | 8        | 2        | 0.1        | 89.84                | 185956           |
| Q, K, V                  | 8        | 4        | 0.1        | 89.98                | 185956           |
| Q, K, V                  | 8        | 8        | 0.1        | 90.25                | 185956           |

Test accuracy comparison across all LoRA configurations and baseline

### Optuna Hyperparameter Search

- **Tool:** Optuna with TPESampler
- **Objective:** Maximize validation accuracy
- **Search Space:** Rank  $\in \{2, 4, 8\}$ , Alpha  $\in \{2, 4, 8\}$ , Dropout fixed at 0.1
- **Number of Trials:** 9
- **Best Configuration Found:** Rank = 4, Alpha = 2, Dropout = 0.1
- **Best Validation Accuracy:** 90.27 %

### Key Observations

- Higher alpha/rank ratio generally leads to stronger adaptation but risks overfitting on limited epochs.
- LoRA significantly increased the number of trainable parameters while keeping pre-trained backbone weights frozen, offering a strong accuracy-efficiency trade-off.
- Optuna converged to an optimal LoRA configuration in faster than exhaustive grid search.

## Q2: Adversarial Attacks using IBM ART on CIFAR-10

### ResNet-18 Baseline Training (Clean CIFAR-10)

| Epoch | Val Acc (%) |
|-------|-------------|
| 1     | 33.21       |
| 2     | 48.84       |
| 3     | 57.22       |
| 4     | 60.46       |
| 5     | 67.10       |
| 6     | 72.69       |

ResNet-18 training on CIFAR-10 ( $\geq 72\%$  test accuracy target (Early stopping))

### FGSM Attack — From Scratch vs IBM ART

## FGSM Attack Results — Accuracy vs Perturbation Strength ( $\epsilon$ )

| $\epsilon$ | Adversarial Acc — Scratch (%) | Adversarial Acc — ART (%) |
|------------|-------------------------------|---------------------------|
| 0.01       | 58.86                         | 67.31                     |
| 0.05       | 22.16                         | 35.77                     |
| 0.10       | 8.55                          | 21.94                     |

FGSM at various perturbation strengths  $\epsilon$

### Key Observations — FGSM

- Both scratch and ART-based FGSM implementations produced visually similar adversarial images.
- Small perturbations were produced which were imperceptible to the human eye yet highly effective at fooling the model.
- Model accuracy drops steeply with increasing  $\epsilon$ , with near-random performance ( $\sim 10\%$ ) observed at higher values of  $\epsilon$ .
- The ART implementation almost matched the from-scratch FGSM outputs, validating both implementations.

## Adversarial Detection — PGD and BIM

### Detection Model Setup

- **Detector Architecture:** ResNet-34 with binary output head (clean = 0, adversarial = 1)
- **Training Data:** Mixed clean + adversarial images (50/50 split)
- **Adversarial Generation:** IBM ART — `ProjectedGradientDescent` and `BasicIterativeMethod`
- **Loss:** Binary Cross-Entropy
- **Optimizer:** Adam

### PGD-based Adversarial Detection

| Epoch | Train Loss | Val Acc (%) |
|-------|------------|-------------|
| 1     | 0.0935     | 99.94       |
| 2     | 0.0029     | 100.00      |
| 3     | 0.0005     | 100.00      |
| 4     | 0.0001     | 100.00      |
| 5     | 0.0000     | 100.00      |
| 6     | 0.0000     | 100.00      |
| 7     | 0.0000     | 100.00      |
| 8     | 0.0000     | 100.00      |
| 9     | 0.0000     | 100.00      |
| 10    | 0.0000     | 100.00      |

PGD adversarial images

## BIM-based Adversarial Detection

| Epoch | Train Loss | Val Acc (%) |
|-------|------------|-------------|
| 1     | 0.0951     | 98.72       |
| 2     | 0.0066     | 99.88       |
| 3     | 0.0022     | 99.94       |
| 4     | 0.0018     | 99.97       |
| 5     | 0.0008     | 99.97       |
| 6     | 0.0015     | 99.91       |
| 7     | 0.0035     | 100.00      |
| 8     | 0.0001     | 100.00      |
| 9     | 0.0000     | 100.00      |
| 10    | 0.0000     | 100.00      |

BIM adversarial images

## Detection Performance Comparison

| Attack Type  | Attack Method | Detection Acc (%) |
|--------------|---------------|-------------------|
| Clean vs PGD | IBM ART PGD   | 100               |
| Clean vs BIM | IBM ART BIM   | 100               |

## Key Observations — Adversarial Detection

- BIM takes more time to converge as compared to PGD
- PGD is a stronger, iterative attack; its adversarial images exhibit more structured perturbations that the detector can exploit as distinguishing features.
- BIM (a variant of PGD without projection) also introduces detectable statistical artifacts, but detection may vary with step size and iteration count.
- The ResNet-34 detector learned to distinguish clean vs adversarial images by picking up high-frequency noise patterns introduced by gradient-based attacks.

## WandB Visualisations

The following results were logged to WandB:

- Q1: Train-val loss & accuracy curves for all 9 LoRA experiments + baseline, class-wise test accuracy histogram, gradient update norms on LoRA weights.
- Q2: 10 sample images (per category) showing:
  - Original clean image
  - FGSM adversarial (from scratch)
  - FGSM adversarial (IBM ART)
  - PGD adversarial (IBM ART)
  - BIM adversarial (IBM ART)

## Links

- **GitHub (Assignment 5 branch):** [https://github.com/arpitadeshmukh/MLOps-Arpita\\_Abhijit\\_Deshmukh-B23CM1007/tree/Assignment5](https://github.com/arpitadeshmukh/MLOps-Arpita_Abhijit_Deshmukh-B23CM1007/tree/Assignment5)
- **WandB Link Q1:** <https://wandb.ai/b23cm1007-indian-institute-of-technology-jodhpur/MLOPS%20Assignment-5/workspace?nw=nwuserb23cm1007>

- **WandB Link Q2:** <https://wandb.ai/b23cm1007-indian-institute-of-technology-jodhpur/MLOPS-Assi-runs/04wc8aso?nw=nwuserb23cm1007>
- **HuggingFace (Best LoRA Model):** <https://huggingface.co/arpita2desh/Vit-Lora-CIFAR100-MLOPS>